DATASCI 406 FA 26, Quiz 3.3

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Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

A line is fit to Lake Huron's level against year, and the residuals are positively correlated at short lags. Resampling single residuals gives an SE for the slope close to the one from `summary`. Resampling blocks of 10 residuals gives about twice that. Why does resampling single residuals understate the SE?

1. A single residual has a smaller variance than the average of a block of residuals, so the noise is too small.
2. Least squares makes the residuals sum to 0, so resampling single residuals understates the noise about the line.
3. The bootstrap series have independent noise, without the runs of high and low years that tilt the real line.
4. With $\ell = 1$ there are fewer distinct bootstrap samples, so the bootstrap distribution is too narrow.

Solution

Answer: 3. Runs of high or low years tilt the line, so the real slope varies more than independent noise says. Single residuals give independent noise, which is what `summary` assumes. Blocks keep the runs. Option 1 has the variances backwards. Option 2 is true but unrelated to dependence. Option 4: the number of distinct samples does not set the SE.

Question 2 (1 point)

Part C regressed daily temperature on a 2012 indicator plus a yearly cycle and block-bootstrapped the residuals. A student leaves out the cycle, regresses temperature on the indicator alone, and block-bootstraps those residuals. The SE of the 2012 effect rises with every ℓ tried, from 0.6 at $\ell = 1$ to 4.5 at $\ell = 90$, and never levels off. What is wrong?

1. The blocks are too short. With a yearly cycle the dependence reaches a year, so ℓ has to be 365 or more.
2. The residuals are not stationary. A January block and a July block sit at different levels and are not interchangeable.
3. The residuals are dependent. The block bootstrap needs independent values, which only the differences between days provide.
4. The residuals are not normal. The block bootstrap needs normal noise, which only the fit with the cycle provides.

Solution

Answer: 2. Blocks handle dependence. What they need is stationarity: a block from any part of the series could have come from any other part. With the cycle left in, January blocks are low and July blocks high, and a longer block carries more of the cycle, so the SE keeps growing. With the cycle fitted, the SE levels off near 0.47 by $\ell = 14$. Option 1: longer blocks make it worse. Option 3 confuses stationarity with independence; blocks exist so that independence is not needed. Option 4: nothing in the method needs normality.

Question 3 (1 point)

For a regression of y on x, the residuals plotted against x spread out as x grows. The slope is a smooth statistic, so each method below gives an SE for it. Which one is misleading here?

1. Cases: draw n rows (x_i, y_i) with replacement, and refit the line to them.
2. Errors: $y_i^* = \hat{y}_i + e_i^*$, with e_i^* drawn from the residuals, and refit.
3. Wild: $y_i^* = \hat{y}_i + e_i V_i$, with $V_i = \pm 1$ at random, and refit.
4. Jackknife: leave out one row at a time, refit, and use the spread of the n slopes.

Solution

Answer: 2. Resampling errors moves residuals from one x to another. The extreme x values have the most leverage, so if the errors are largest there the SE is understated, and if smallest, overstated. Cases, wild and the jackknife never move a residual away from its x. Plot the residuals against x before resampling errors.

Question 4 (1 point)

The interquartile range of the high school graduation rate across the 50 states is bootstrapped two ways. Resampling states within each of the four census regions, keeping each region's count, gives an SE of 2.4. Resampling all 50 states gives 3.0. Why is the second SE larger?

1. The count from each region varies from sample to sample, and the regions differ.
2. Each region is a smaller pool, so within-region samples repeat states more often.
3. The IQR is not smooth, and the plain bootstrap overstates the SE of such statistics.
4. States within a region are alike, so the within-region spread is biased downward.

Solution

Answer: 1. The South is low and the West high. Resampling all 50 varies the count from each region, which the data never do, and that adds spread. Resampling within regions keeps the counts fixed. Option 2: repeats are how every bootstrap sample is built. Option 3: the IQR is smooth enough; the failure was the max. Option 4: there is no bias, and within-group resampling matches how the data arose.

Rubric. 1 pt per question, no partial credit.